

Hatchet Interaction Framework

Project-Hatchet/Interaction-Framework · branch `fix/knob-first-use` @ `22fec56` · ~2,600 lines of SQF/config across 4 addons

What it is

A CBA-based Arma 3 mod that lets vehicle authors bind scripts and 3D cockpit interactions (buttons, levers, knobs) to vehicles through config alone. Structure is clean and idiomatic HEMTT/CBA:

addons/main

Prefix, version, and macro plumbing. Boilerplate only.

addons/core

Module lifecycle: per-seat config discovery, load, perFrame / perSecond / perFixed loops, shutdown.

addons/interaction

The product: Draw3D-driven cockpit interaction, pointing system, mouse-blocker display, CBA settings, modded keybinds.

addons/util

Grab-bag: PID controller, in-vehicle salute, countermeasures from other seats, turret zoom/reload, PiP toggle, coordinate-finder dev tool.

The recurring bug class: globals that stay undefined until first use, read through non-lazy `&&` or bare `if`. Commit `22fec56` fixed exactly this in `knobAnimate` / `dragStart` — findings 3, 4, 5, and 6 below are more members of the same family. Cheapest systemic fix: initialize *every* interaction-state global in `fnc_init`, and brace every second operand that touches possibly-nil state.

High — broken code paths

HIGH 1. `hct_core_fnc_start` can never start a module

`core/fnc_start.sqf:16` — `private ["_moduleName", ...]` appears *after* `params`, resetting the `_moduleName` parameter to nil, so the `_moduleName == _name` comparison errors and no module ever matches.

Fix: drop `_moduleName` from the private list (and declare `_module` / `_name` properly).

HIGH 2. Undefined `_projectPrefix` in `fnc_load`

`core/fnc_load.sqf:24-25` — the `perFixed` and `Draw3D` default paths use `(_projectPrefix + "...")`, but `_projectPrefix` is a local that only exists in `fnc_loadAll`. This is the path that registers the built-in `interaction` pseudo-module on every vehicle entry (`startLoops:18`), so it throws an undefined-variable error each time.

Fix: read `_vehicle` `getVariable ["hct_projectPrefix", ""]`, or use the `"hct_%1..."` literal like the neighbouring lines. Also make `_result` (line 31) private.

HIGH 3. Debug setting variable name mismatch

Three sites read `hct_interactionShowDebugMessages` (`fnc_drag.sqf:28,41`, `fnc_buttonUp.sqf:25`) but the CBA setting registers `hct_interaction_showDebugMessages` (`fnc_preInit.sqf:115`). The read is always nil: the debug setting can never work, and `if (nil)` throws in the drag-update and knob-click paths.

Fix: rename the three call sites to the registered setting name.

HIGH 4. `hct_interaction_pointStart` is never initialized

`fnc_draw3D.sqf:39` reads it every frame, but it's only assigned inside `pointStart` / `pointCalculate` — and nothing in the repo calls `pointStart`. On a clean install the Draw3D handler hits an undefined global in `if`, which also cuts off the knob-release logic and module Draw3D callbacks below it.

Fix: `hct_interaction_pointStart = false;` in `fnc_init`.

HIGH 5. First knob hover repeats the just-fixed first-use crash

`fnc_drawLabel.sqf:64,67` — non-lazy `&&` evaluates `!hct_interaction_scrolledHolding`, which is undefined until line 124 has run once; line 124 is unreachable while no interaction is hovered (early exit at line 25). If the first interaction hovered in a session is a knob, drawLabel errors — the same family as the 22fec56 fix.

Fix: initialize `hct_interaction_scrolledHolding = false;` in `fnc_init` (and/or brace the second operands).

HIGH 6. Util keybinds error outside HCT vehicles

`util/inputAction.inc.sqf:11,18,22` — `hct_vehicle unitTurret hct_player` runs with `hct_vehicle` nil whenever the player is on foot in first person (camera "INTERNAL") or in a vehicle without HCT config, throwing on every countermeasure/zoom keypress.

Fix: guard each handler with `if (isNil "hct_vehicle") exitWith {}`.

Medium — wrong behavior

MED 7. Door animations misread in `getAnimPhase`

`core/fnc_getAnimPhase.sqf:17-21` — a missing `else` means `_type = ANIMTYPE_ASRC` unconditionally overwrites `ANIMTYPE_DOOR`; the `doorPhase` branch is unreachable, so door-source animations read the wrong phase. `_type` is also not private.

MED 8. Levers pass the wrong argument to `interactionCondition`

`fnc_leverAnimate.sqf:17` calls the condition with `_this` (the full 8-element argument array); `buttonDown` / `knobAnimate` pass the vehicle. A condition written per the README (`damage _this < 0.5`) works on buttons and errors on levers.

MED 9. knobAnimate animates before its own gating checks

`fnc_knobAnimate.sqf:18-31` — `animateSource` runs at line 23, but `interactionCondition` and the `hct_animating_keys` re-entrancy check come after (lines 30–31): a condition-disabled knob still physically moves while its script hooks are skipped. Line 16 also compares the held interaction's *class name* (`knobHolding # 0`) with the *animation name* — only correct when the config class happens to be named after the animation.

MED 10. Inconsistent dragging-callback contract

`fnc_drag.sqf:34,47` passes `[vehicle, animationName]` to the user's `dragging` code; `fnc_draw3D.sqf:58` and the README use `[vehicle, animationPhase]`. Downstream code can't satisfy both.

MED 11. perFixed/perSecond spawn-and-sleep scheduling

The `perFixed` PFH fires every 0.02 s (`startLoops:33`) and each tick `spawn`s a script that sleeps $1/(n+1)$ s between modules (`fnc_perFixed.sqf:28-39`). Steady state is dozens of concurrent scheduled scripts, and "fixed-interval" callbacks actually run at the scheduler's mercy, up to ~1 s late. The README documents 0.2 s. `perSecond` has the same shape.

Fix: call module functions directly in the PFH (no spawn/sleep); stagger with per-module offsets if load-spreading is the goal.

MED 12. Knob/drag state survives vehicle changes

Interaction's `fnc_shutDown` is an empty stub and nothing resets `hct_interaction_knobHolding`, `hct_animating_keys`, or `hct_interaction_currentButton` on exit. Leave a vehicle mid-knob-animation and `draw3D:44-59` runs the old knob config against the new vehicle, eventually firing `dragStop` with the wrong vehicle.

MED 13. Shutdown edge cases with nil handles

`core/fnc_shutDownAll.sqf:16-18` passes possibly-nil PFH handles to CBA removal; `fnc_perFrame.sqf:21` passes possibly-nil `hct_vehicle` into `shutDownAll`, whose `getVariable` then errors when both are nil (death / vehicle-destroyed paths).

Low — polish and drift

LOW 14. Leftover hot-path diag_log

`22fec56` cleaned `knobAnimate/dragStart` but the same unconditional logs remain in `buttonDown:14`, `buttonUp:14`, `leverAnimate:15`, `dragStop:11`, and `drag:13` — the last runs every frame while a knob is held.

LOW 15. Conditions recompiled in the hot path

`fnc_checkInteraction.sqf:29` recompiles each group's `condition` string on every update pass; press handlers recompile `interactionCondition` per press. `loadItem` already precompiles everything else — compiling these once there (with a `{true}` default) also fixes the subtle case where an *empty* condition yields `call {} → Nothing → ! errors`.

LOW 16. animSpeed default is "instant", README says 1

`fnc_loadItem.sqf:75,99` turns unset `animSpeed` into boolean `true` — the `animateSource` instant flag — so unconfigured levers/knobs snap instead of animating at speed 1.

LOW 17. Examples/Interactionconfig.cpp is stale

Uses subclass `vtx` where the framework looks for `hct*; condition="engineOn"` is invalid SQF (needs `isEngineOn _this`); `type / code / scrollSteps / dragLooping` aren't read by the loader (it wants `scrollIncrement / looping`). This is the first file an adopter copies.

LOW 18. Turret config granularity

`core/fnc_getVehicleConfig.sqf:35` uses only `_turretIndex # 0`, so sub-turrets `[0]` and `[0,0]` both map to `hct_turret_0`; only the first copilot turret is considered (line 29).

LOW 19. Docs and cosmetics

- README badge says 0.2.0; CHANGELOG is at 0.3.1. README ends mid-sentence ("The above shows").
- `animStart / animEnd` actually receive `[vehicle, animation, label, target]`, not the README's `[vehicle] / [vehicle, label]`.
- Keybind tooltips for "Previous/Next Setting" are swapped (`CfgUserActions.hpp`).
- Placeholder authors "Your Name", "Other Author" in interaction and util `config.cpp`.
- `fnc_shutDownAll` header comment names the wrong function; `hasModule` leaks `_name`; `pointDraw` deletes from the array it's iterating (one-frame draw glitch).
- `leverAnimate` writes with `animateSource` but polls with `animationPhase` — only correct when model-anim and source names match; `core/fnc_getAnimPhase` exists to resolve exactly this and isn't used.

Suggested order of attack

STEP	CHANGE	WHY FIRST
1	Init all interaction globals in <code>fnc_init</code> ; lazy-brace nil-touching <code>&&</code>	Kills findings 4, 5 and the whole recurring bug class
2	Fix <code>_projectPrefix</code> in <code>fnc_load</code> + debug-var rename + <code>fnc_start</code> private list	Three mechanical one-liners, all High

3	Guard util keybind handlers on <code>hct_vehicle</code>	Stops RPT spam for every player on foot
4	Precompile conditions in <code>loadItem</code> ; unify condition/callback arguments	Perf + fixes lever/drag contract drift in one pass
5	Rework perFixed/perSecond scheduling; clear knob state in shutdown	Behavioral, needs in-game testing
6	Refresh README + Examples	Adopter-facing accuracy